PHISHING EMAIL DETECTION & AWARENESS REPORT

INTRODUCTION

Phishing is a type of cyber attack where attackers send fraudulent emails pretending to be from legitimate organizations. The goal is to trick users into revealing sensitive information such as usernames, passwords, or banking details. Phishing emails often contain malicious links or attachments designed to steal data.

Phishing Email Sample

For this analysis, a phishing email sample pretending to be from PayPal was examined.

Example message:

“Your PayPal account has been temporarily limited. Please verify your account immediately by clicking the link below to avoid suspension.”

← Service@paypalsecurityteam.com

Your account has been temporarily limited Inbox

PayPal

Dear customer,

⚠️ Your account has been **temporarily limited!**

We noticed some unusual activity in your account. **Please verify** your account information by clicking the button below to avoid getting your account **suspended**.

Verify Your Account

If you do not verify your account within 48 hours, it will be permanently suspended.

To help you restore your account, **log in** and follow the instructions.

Thanks,
PayPal

Help | Security | Privacy

If you need assistance, please log in to your PayPal account and click Help in the top right corner of any PayPal page. Please don't reply to this email. We send these emails to keep you informed of possible fraudulent activity in your account and to protect your account.

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Suite 9-11 11 Royce Road, Onoteu City,

Victoria, 3000 Australia

PP ID EXAMPLE

Phishing Indicators

The following indicators suggest that the email is a phishing attempt:

- Suspicious sender email address
- Fake domain name
- Urgent message requesting immediate action
- Suspicious hyperlink redirecting to an unknown website

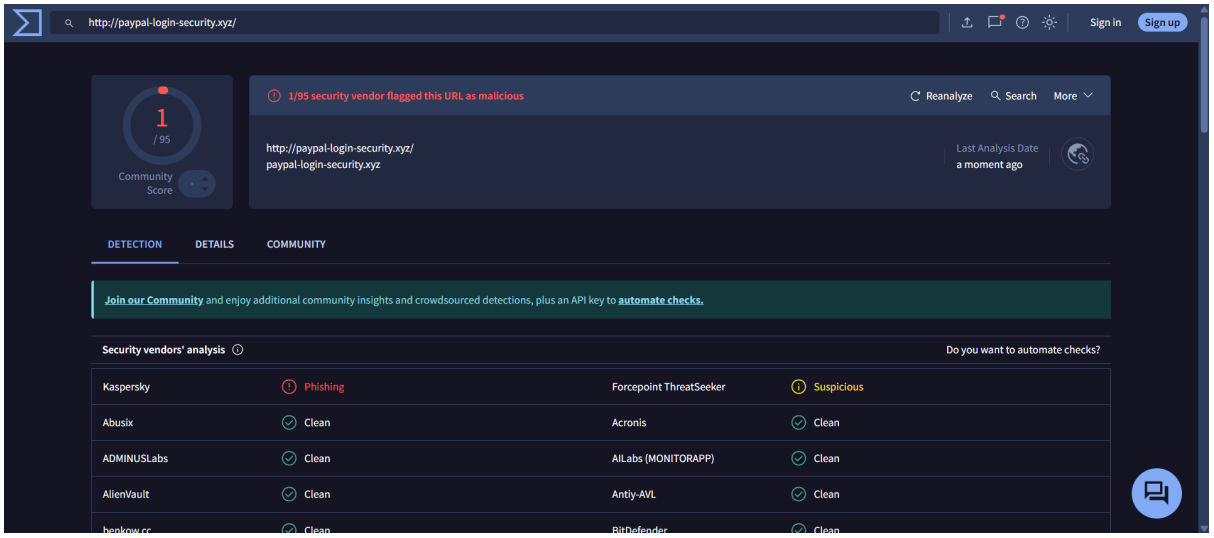
These characteristics are commonly used in phishing attacks to manipulate users.

Link Analysis

The suspicious link from the email was analyzed using VirusTotal.

Results:

- Several security vendors flagged the URL as malicious.
- The domain was identified as suspicious and not associated with the legitimate company.



Domain Analysis

The domain was further analyzed using MXToolbox.

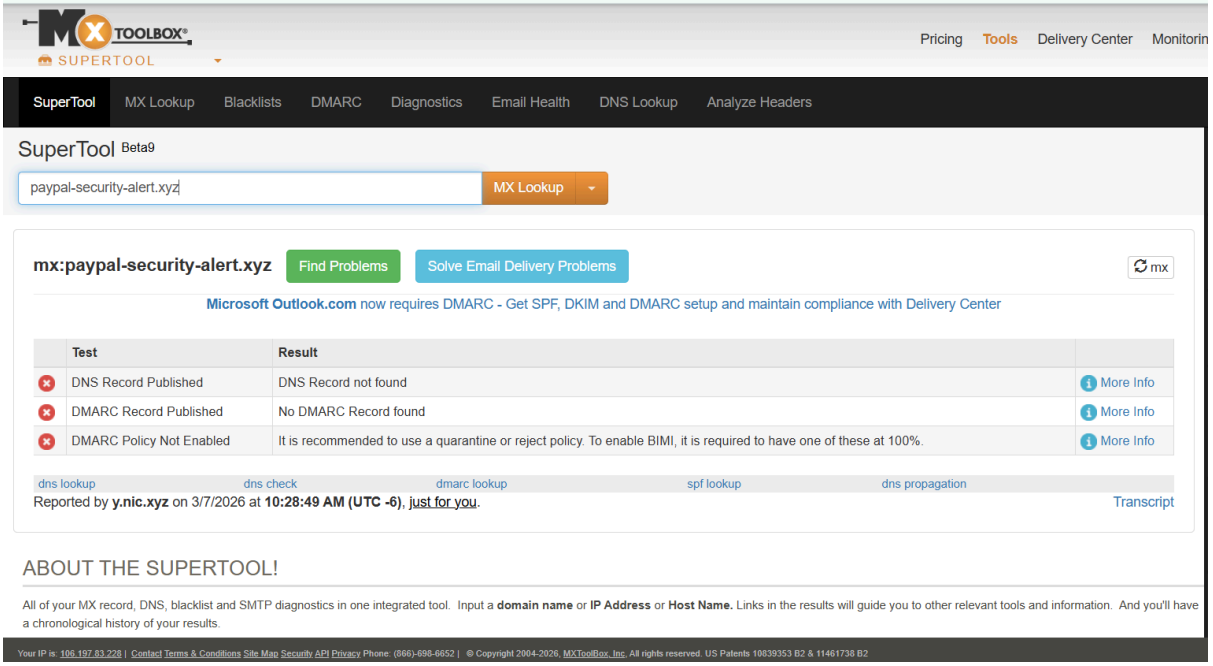
Findings:

- The domain appears newly created.
- It does not belong to the official PayPal domain.
- This suggests the domain was likely created for phishing purposes.

The domain paypal-security-alert.xyz was analyzed using MXToolbox.

The analysis shows that the domain does not have proper DNS and DMARC records configured. The absence of these security records

indicates that the domain may be suspicious and potentially used for phishing activities.



Email Classification

Based on the analysis,

The email is classified as a Phishing email because it contains a suspicious domain, missing DNS security records, and a potentially malicious link.

EMAIL	CLASSIFICATION	REASON
Fake PayPal email	Phishing	Fake domain + suspicious link

Phishing Technique Used

The attacker used the following phishing techniques:

- Email Spoofing – pretending to be a legitimate company
- Social Engineering – creating urgency to force the user to act quickly
- Credential Harvesting – directing users to a fake login page to steal credentials

Prevention Guidelines

Users can protect themselves from phishing attacks by following these security practices:

- Always verify the sender's email address
- Avoid clicking unknown or suspicious links
- Check website URLs carefully before entering credentials
- Enable two-factor authentication (2FA)
- Report suspicious emails to the organization's security team

Conclusion

Phishing attacks remain one of the most common cyber threats. By carefully analyzing email indicators, links, and domains, users can identify phishing attempts and avoid becoming victims. Security awareness and proper verification practices are essential to prevent phishing attacks.